

litemall — Upstream Parity Audit & Wave-4 Design Plan

2026-07-13 · Prepared before any implementation, per plan-first workflow · Sources: full controller inventories of upstream `linlinjava/litemall` (litemall-wx-api, litemall-admin-api, litemall-core kept in-repo as reference) and `crmeb_java` (admin/service/common/front), inventory of the current microservices, and a 12-agent design+adversarial-review workflow grounded in both codebases.

Purpose. (1) Establish how much of upstream litemall and crmeb_java the current `/srv` microservices already cover; (2) design the remaining gaps — including the crmeb verticals selected for adoption (stores/write-off, DIY page builder, article CMS, freight templates, order export, Yly printer + OnePass as DDD seams) — so the legacy `litemall-wx-api` and `litemall-admin-api` modules can be deleted. Every design below was adversarially verified against the actual code; review corrections are already folded in.

1 · Executive summary

- After Waves 2-3, the microservices cover **~90% of upstream litemall** (customer surface nearly complete; admin close) and **all of crmeb's core commerce** (product/SKU, orders, coupons, seckill, bargain, group-buy, points, levels, sign-in, recharge, funds) except distribution/brokerage.
- The single biggest deletion blocker: **no customer registration / password reset / profile endpoint exists** — the shop runs on one seed user.
- Wave 4 closes every remaining litemall blocker and adopts seven crmeb verticals, in **six designed work packages** spread over the existing four active worktrees (promotion is parked).
- Key discovery from adversarial review:** the repo already carries crmeb-shaped freight-template tables (V8) and `litemall_goods.temp_id, weight, volume` columns (V2) from an earlier wave — Wave 4 reuses them instead of duplicating.
- Out of scope by explicit decision: WeChat ecosystem (replaced by dual-JWT + wallet + PSP seam), brokerage/distribution (deferred), user tags/groups, live streaming.
- After Wave 4 merges: remove the `customer-wx-api` route, drop `litemall-wx-api` + `litemall-admin-api` from the root pom. **Keep litemall-core** (order uses its NotifyService; express/storage seams reused) and `litemall-db`.

2 · Part I — Parity audit

2.1 Covered today (no action)

Vertical	Where	Notes
Catalog / search / detail / videos / comments	goods-management	OCS-backed facets, typo-tolerance, ranking signals — ahead of upstream
Cart, checkout, orders, timeline, refunds	order	Single-transition-source state machine + timeline + outbox; tender-parity refunds
Coupons, group-buy, seckill, bargain, campaigns/RFM	promotion-service	Customer + admin surfaces live
Aftersale/RMA	order	Apply → admin approve/reject → tender-parity refund
Collect / footprint / feedback / comment-post	goods-management	+ admin lists
Wallet (balance/recharge/extract/bills), loyalty (points/sign-in/levels)	order (/srv/wallet), loyalty-service	Exceeds litemall; matches crmeb integral/level/sign verticals. <code>litemall-wallet-service</code> module is an empty placeholder.

Scheduled lifecycle tasks	order, promotion, goods	Unpaid auto-cancel, auto-confirm, CJ sync, outbox relay,groupon expiry, catalog refresh — verified present
Admin CRUD (goods/category/brand/keyword/FAQ/storage/users/orders/comments) + dashboards	goods-management, order	
Admin accounts, roles, notices, ads, audit log	gateway-admin edge controllers	
CJ dropshipping end-to-end	goods-management + order	Sourcing, videos, warehouse, live stock, createOrderV2 lifecycle, tracking, disputes, balance — no upstream equivalent

2.2 Gaps closed by Wave 4

Gap	Origin	Wave-4 package
Registration / password reset / profile	litemall wx-api	Account self-service (§3.6)
Region cascade, customer upload, topic admin CRUD, search history, comment reply, system config, admin profile/notices	litemall wx/admin-api	Small parity bundle (§3.5)
Shipping freight templates	crmeb	Freight templates (§3.1)
Physical stores + pickup + write-off (核销)	crmeb	Stores & write-off (§3.2)
Article CMS + DIY page builder	crmeb	Content (§3.3)
Receipt printer (Yly), OnePass, local-order tracking	crmeb + litemall core	Fulfillment seams (§3.4)
Order Excel/CSV export, offline mark-paid, channel stats, batch aftersale	crmeb + litemall admin-api	Small parity bundle (§3.5)

2.3 Explicit non-goals

WeChat pay/login/messages/QR (replaced by design) · brokerage/distribution (deferred — the only remaining crmeb feature gap after Wave 4) · user tags/groups CRM · live streaming · SMS flows (seam exists in litemall-core; no sender configured) · login kaptcha.

3 · Part II — Wave-4 designs (adversarially verified)

Each package was designed by an agent that read both the crmeb reference implementation and the current services, then independently verified by an adversarial reviewer who re-checked every load-bearing claim against the code. Verdicts were "needs-fixes" across the board; all corrections are incorporated below and in the CLAUDE.md tasks.

3.1 Shipping freight templates → **order** **goods-management** **both gateways**

Placement. FreightTemplate aggregate + resolution service + admin CRUD in `litemall-order` (freight is consumed only by freight-quote and submit; the address book is order-owned). Goods only declare a binding.

Review discovery. V8 already created `litemall_shipping_templates{,_region,_free}` and V2 already added `litemall_goods.temp_id / weight / volume` — the design *reuses* them: one ALTER migration adds `country_code` + normalized `province_name` (addresses store free-text international names; the Chinese region tree is unusable for matching), and goods-management merely hand-maps the existing `temp_id` column (0 = unbound). `FlywayMigrationTest` must be updated.

Resolution ladder (single authority for quote AND submit): global free-shipping min → template group rules (crmeb appoint/first/continue formula, most-specific region match, free-rule OR semantics) → flat fallback. Combine mode `max` by default (crmeb sums — a silent price hike on multi-template carts; config-flippable). CJ carts skip templates. Documented deviation: crmeb ships appoint=1-unmatched *free*; we flat-charge. Feature flag `freight.template.enabled`; quote never 5xxs for freight reasons.

Endpoints. `POST /srv/order/freight-quote` gains optional owner-scoped `addressId` + `breakdown[]` (X-User-Id stays optional — anonymous quotes must not break); admin `/srv/private/admin/freight/**`: template CRUD (delete refused while referenced), set-default, selectlist (bare array), `/preview` dry-run.

3.2 Physical stores + pickup + write-off → **order** **both gateways**

Placement. Entirely in `litemall-order`: the verify code is order state and write-off is a state transition that must commit with the timeline hop. New `litemall_store` table + 5 columns on `litemall_order` (`delivery_type/store_id/verify_code` UNIQUE/`verify_time/verified_by`). Verify code as a column (crmeb parity; exactly-once via one conditional UPDATE — the `markPaidIfCreated` trick).

Flow. Submit accepts pickup + storeId (+ name/mobile): freight 0, address optional, CJ lines refused, *all validation 422s in the orchestrator pre-check zone* (the in-TX-422→502 landmine, caught by review). Code generated at pay (deviation from crmeb-at-create: unpaid orders never carry a redeemable code). Write-off: admin preview → transactional commit through the orchestrator (review correction: bare service call had no TX) → PAID→DELIVERED with timeline hop; three distinct 422s; double-scan loses cleanly. Ship refuses pickup orders. Staff RBAC deferred (crmeb binds wx-customer uids — different identity universe); write-off is `ROLE_ADMIN` with `verified_by` audit.

Review fixes. `litemall_order.address` is `varchar(127)` → store only `"PICKUP: <store.name>"`; PAID→DELIVERED widening is actually guarded by the handle-option gate + conditional UPDATE (rationale corrected); order list DTO gains `deliveryType` so My-Orders tabs can relabel; `/srv/store/**` added to public-paths.

3.3 Article CMS + DIY page builder (phase-1) → **goods-management** **both gateways**

Placement. A `content` subdomain in goods-management (it owns topics/FAQ/home/storage already). Three new tables: article category (declared deviation — crmeb reuses its shared category tree), article (sanitized MEDIUMTEXT content, atomic `view_count`, `hidden|published`), page (palette-v1 JSON ≤64KB, `draft|active`, *schema-enforced* single active home via generated-column unique index — review caught that `FOR UPDATE` on zero rows serializes nothing).

Palette v1 (normative committed spec): ordered component array — banner, image-row, goods-list (`byIds|byCategory|hot|new`), coupon-strip, seckill-strip, article-strip, rich-text — storing IDs/queries, *not* crmeb's stale denormalized product snapshots. Strips resolve *client-side*: promotion down ⇒ strips hidden, never a broken home. Review corrections baked into the spec: promotion endpoints take no limit param and return bare arrays; `/srv/goods/batch` returns a raw Map; hot/new modes serve local-only goods (divergence from the OCS-boosted legacy home, documented). Rich text sanitized with a custom jsoup Safelist, clean-and-store semantics.

Rollout. No active home seeded — today's home stays pixel-identical until an admin activates a page (draft "Default Home" seeded). Admin editor is a structured per-component form (from `GET /palette`), not drag-drop. Customer: `/srv/article/*`, `/srv/page/home`, `/srv/page/{id}`; `errno 642` triggers the SPA legacy-home fallback.

3.4 Receipt printer (Yly) + OnePass + local tracking → order both gateways

Shape. Two ports behind the module's facade convention, disabled by default (promotion Matomo pattern), *zero migrations*: `ReceiptPrinterPort` (no-op logging adapter = the live-verification surface; Yly adapter with OAuth token cached ~30d, `origin_id=orderSn` — fixes crmeb's hardcoded "order111", giving vendor-side exactly-once) and `ExpressQueryPort` (noop | kdniao wrapping litemall-core's `ExpressService` | OnePass v2/expr/query — verified call shape). Auto-print rides `@TransactionalEventListener(AFTER_COMMIT)` on order-paid → async → catch-all: a printer outage can never block or fail a payment (improvement over crmeb's synchronous call).

Tracking. `CjTrackingService` → `OrderTrackingService`; local shipped orders consult the port: disabled → clean payload + `note:"tracking provider disabled"`; enabled → `events[]`; CJ branch byte-identical. Never 5xx, never a hang. Review findings: the customer SPA has *no* tracking view today (gateway-api builds one); kdniao enablement fights core's profile.yml config precedence + Chinese vendor names → OnePass is the practical provider; print dispatch gets a small dedicated executor.

New admin endpoints. `POST .../print-receipt` (641 disabled / 642 failed) + `GET .../fulfillment/config`. OnePass SMS explicitly deferred to litemall-core's existing `SmsSender` seam.

3.5 Small parity bundle (9 thin verticals) → goods-management order gateway-admin

Zero new tables, zero migrations — every item maps onto an existing hand-maintained table (verified live against the dev DB).

#	Item	Owner	Essence
1	Region cascade	goods-mgmt	<code>/srv/region/list?pid=</code> + <code>/clist</code> tree, in-memory cache; public-paths entry required (review); data is China-only → SPA cascade optional (R1)
2	Customer upload + public fetch	goods-mgmt	<code>POST /srv/storage/upload</code> (X-User-Id, 5MB, magic-byte whitelist) + <code>GET /srv/storage/fetch/{key:.+}</code> ; fixes <code>storage.local.address</code> pointing at a nonexistent route (review: no fetch route ever existed)
3	Topic admin CRUD	goods-mgmt	AdminBrand template; litemall-db service surface already complete
4	Search history	goods-mgmt	Write on search (soft-fail, dedupe via new latest-row query — review), <code>clearhistory</code> , admin list
5	Comment reply	goods-mgmt	Sets <code>admin_content</code> once; re-reply → errno 622 (review: not 620); customer DTO already emits it
6	Order CSV export	order	Streamed CSV (BOM, RFC-4180), list filters + start/end, keyset pagination + lean projection (review), 10k cap + TRUNCATED trailer; queries live in litemall-db db.dao (review: no module-local mapper pattern exists)
7	Order admin extras	order	Offline mark-paid (<code>pay_id OFFLINE:</code> , timeline hop, CJ replay fires — live-fire, confirm dialog names it); <code>stat/channel</code> (closes recorded follow-up); aftersale batch approve/reject (per-id TX, partial-success envelope)
8	System config	gateway-admin edge	<code>/config/{mall,express,order}</code> over <code>litemall_system</code> ; prefix guard + posted keys validated against existing rows (review: <code>updateConfig</code> silently no-ops unknowns); runtime consumption = handoff note
9	Admin profile	gateway-admin edge	Own-password change (BCrypt, 605) + per-admin notice inbox, JWT-adminId-scoped (fixes legacy stub)

3.6 Customer account self-service → gateway-api

Placement. In gateway-api's existing edge auth package (it owns the customer-identity context: credentials, JWT, refresh tokens). One migration: `email` column + hashed single-use reset-token table (V15 pattern). Review: the username unique index already exists — `DuplicateKey`→704 mapping closes the race; mobile `' '` treated as absent for 705.

Endpoints. `POST /auth/register` (policy 8-72 bytes, auto-login returning login's exact shape — the SPA thunk already expects it), `POST /auth/reset` (old→new, revokes all refresh tokens — the guaranteed zero-dependency path), optional email flow behind `reset-mail.enabled:false` (anti-enumeration always-200; 701/703 semantics; `ResetMailSender` port with no-op default — core's `NotifyService` can't mail arbitrary customers), `POST /auth/profile` + `GET /auth/me` (replaces the dead `/srv/user/index` stub). Wallet/loyalty need *no provisioning* — verified both null-coalesce to zero off the user row. Ends the single-seed-user era; register-gift coupon deferred to promotion.

4 · Part III — Execution

4.1 Worktree assignment matrix

Worktree	Wave-4 packages	Migrations owned
order	Freight templates (A) · Stores/pickup/write-off (B) · Export + admin extras (C) · Printer/express seams + tracking (D)	2 (freight ALTER; store/pickup)
goods-management	Parity items 1-5 · Article CMS + DIY pages · freight templd surface	1 (content tables)
gateway-api	Account self-service (+migration) · pickup checkout + /srv/store route · DIY home renderer + articles · customer tracking view · freight addressId · region cascade + ImageUploader + seller response	1 (user email + reset tokens)
gateway-admin	Config + profile edge controllers · freight/store/write-off/article/page admin SPA · tracking panel widening + print button · export/offline-pay/stat-card/batch-aftersale/topic/history/reply SPA	0
promotion	Parked — no Wave-4 assignment (follow-up noted: register-gift coupon)	0

4.2 Dependency order & cross-cutting risks

- **Start first, in parallel:** order (A-D), goods-management (all), gateway-api (account self-service — independent). Gateway SPA work consumes committed handoff specs (same docs/ pattern as Waves 1-3) and can start against them.
- **Flyway:** up to 4 new migrations across 3 worktrees on the shared dev DB. V33 is the last *known*; every implementer checks `flyway_schema_history` immediately before first boot and claims the next free number (V32 was stolen mid-flight once; never `flyway repair`). Whichever service boots first applies shared migrations.
- **litemall-db coordination:** freight/store/user/content changes all hand-edit the shared litemall-db (entities + mapper XMLs — never regenerate; the `now_money / OrderMapper.xml` silent-failure incidents). Rebuild + reinstall to `~/m2`, restart every dependent, verify the nested `B00T-INF/lib` copy in running `exec jars`; concurrent `-am` builds overwrite the shared jar.
- **svcsecurity:** new anonymous paths (`/srv/region/**`, `/srv/storage/fetch/**`, `/srv/article/**`, `/srv/page/**`, `/srv/store/**`) must be added to `public-paths` — `deny-by-default` catches them otherwise.
- **Acceptance through gateways** (:9000/:9001→:8090, :18080) — machine-token ~10-min TTL makes direct service curls flaky.

4.3 Legacy decommission checklist (after Wave 4 merges)

1. Verify no SPA/gateway path still targets `/wx/**`; remove the `customer-wx-api` route (→ :8084) from gateway-api.
2. Drop `litemall-wx-api` and `litemall-admin-api` modules from the root `pom.xml`; archive their reference value in git history.
3. **Keep** `litemall-core` (`NotifyService/ExpressService/StorageService` are live seams) and `litemall-db`.
4. Optionally delete the empty `litemall-wallet-service` placeholder module (wallet lives in order).
5. Decide the one remaining crmeb gap: brokerage/distribution (spread binding at register + commission ledger + extract approval — fits `promotion-service` + `wallet extract`; a future wave).

4.4 Key open decisions (recommended defaults chosen in the designs)

Decision	Default
Freight multi-template combine	<code>max</code> (crmeb-exact <code>sum</code> behind config)
By-weight/volume freight	Defer — columns exist, entities unmapped; piece-only v1
Pickup status model	Reuse PAID (no <code>READY_FOR_PICKUP</code>); <code>verify-code</code> at pay, no expiry this wave
DIY home replaces legacy home?	No — fallback until an admin activates a page
Express provider	OnePass over <code>kdniao</code> (config precedence + Chinese vendor names make <code>kdniao</code> impractical without a ship-dialog vendor dropdown)

Offline mark-paid on CJ orders	Fires the CJ replay (live-fire; confirm dialog names it; sandbox flag available)
Email reset flow	Shipped but disabled by default; old→new change is the guaranteed path
Region data (China-only)	Ship for parity; SPA cascade optional; international seeding is a separate data task

Method: 4 inventory agents (upstream wx-api+core, admin-api, crmeb, current services) → gap matrix → 6 design agents + 6 adversarial verify agents (each grounded in the real code; every "needs-fixes" verdict's corrections folded in) → this synthesis. Full per-vertical designs and reviews are preserved in the session scratchpad; the CLAUDE.md Wave-4 blocks are the implementable form of this plan.